Guided Project Proposal: “Exploring Polygons through Programming”

Project Overview

This project introduces students to geometric concepts by using programming to explore and analyze polygons. Students will design a program capable of identifying, calculating, and displaying key properties — including **interior angles, area, height, and perimeter** — for a variety of polygons.

The polygons to be covered include:

- **Quadrilaterals:** square, rectangle, parallelogram
- **Triangles:** equilateral, isosceles, scalene
- **Other polygons:** pentagon and hexagon

Learning Objectives

By completing this project, students will:

- Understand geometric formulas and their computational representations.
- Apply object-oriented programming to model polygon types.
- Perform input validation and handle user interaction through functions or classes.
- Visualize relationships between different polygon properties.

Project Description

Students will create a program (in Python, Java, C++, or another instructor-approved language) that allows a user to:

1. Choose a polygon type.
2. Input the necessary side lengths, base, and height (as applicable).
3. Compute and display:
    - **Perimeter:** Sum of all side lengths.
    - **Area:** Computed using geometry-specific formulas (e.g., $A = b \times h$ for rectangles, or Heron’s formula for scalene triangles).

- **Interior angle:** Derived using the formula $(n-2) \times 180/n$.
- **Height:** Calculated where relevant or input by the user.

Step-by-Step Guidance

1. **Phase 1 – Design:**
    - Define classes or functions for each polygon type.
    - Determine which attributes and methods are shared (e.g., perimeter, interior angle).
    - Research the Math to determine which formulas to use (Heron's, etc..)
2. **Phase 2 – Implementation:**
    - Code the polygon classes or functions.
    - Implement formulas for area, angle, height, and perimeter.
    - Round formula results to 2 decimal points.
3. **Phase 3 – Testing:**
    - Validate calculations with known geometric examples.
    - Include edge case testing (e.g., identical side lengths or zero values).
4. **Phase 4 – Enhancement (Optional):**
    - Add a GUI or visualization using simple graphics.
    - Expand to allow irregular polygons or user-defined sides.